

ADAM JEMMALI

Computer Vision | Sports and EdTech Builder | Agentic AI Systems

Ottawa, ON | Canadian Citizen | 819-968-4495 | adam.official.514@gmail.com | [LinkedIn](#) | [GitHub](#) | [mentrixia.one](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Computer Vision and AI Systems engineer with hands-on production experience building real-time sports analytics pipelines, adaptive EdTech platforms, and multi-agent AI systems. Founder of Mentrixia, a live gamified tutoring SaaS, and Co-Founder of Solen AI, an autonomous software deployment platform. Proficient in Python, PyTorch, FastAPI, and cloud-native distributed systems. Seeking a Summer 2026 internship in software engineering, computer vision, machine learning, AI/ML engineering, full-stack development, backend engineering, data engineering, or applied research across sports technology, educational technology, fintech, defence, cloud infrastructure, or any domain deploying AI systems at production scale.

TECHNICAL SKILLS

Computer Vision: Python, OpenCV, MediaPipe, YOLOv8, CNNs, ResNet, Vision-Language Models (VLMs), Object Detection, Image Segmentation, LoRA Fine-Tuning, ONNX, PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, MLflow, Weights and Biases, model fine-tuning, data augmentation, NumPy, pandas

AI, Agentic Systems, LLMs: LangChain, LangGraph, RAG, multi-agent orchestration, adaptive learning AI, intent classification, self-healing loops, tool-calling LLMs, Anthropic Claude API, Google Gemini 2.5, Groq Llama 3.1, DeepSeek AI, OpenRouter, prompt engineering, reproducible experimentation, ablation studies

System Design: Horizontal and vertical scaling, load balancing, stateless service design, CAP theorem, CP/AP tradeoffs, CQRS, cache invalidation (TTL, write-through, write-behind), message queue architecture, circuit breakers, microservices, multi-region failover, rate limiting, eventual and strong consistency, read/write optimization

Languages: Python, TypeScript, Go, C++, Java, SQL, Bash

Full-Stack and Production: Next.js App Router, Python FastAPI, Tailwind CSS, Supabase (Postgres, Auth, Realtime, Storage), Drizzle ORM, AWS EC2 and S3, Vercel, Stripe Connect, Anthropic Claude API, Postgres FTS, Resend, PostHog, Vercel Analytics, Sentry APM, Markdown/MDX, GitHub Actions CI/CD, Playwright, Vitest, WebRTC, WebSocket, React, REST APIs

Infrastructure and Data: Docker, Kubernetes, Helm, AWS Lambda, SQS, ECR, Jenkins, NATS/JetStream, RabbitMQ, Redis, ChromaDB, Pinecone, MongoDB Atlas, Snowflake, SQLAlchemy async, Alembic, Linux, n8n, Make.com, Twilio, ElevenLabs, PyTest, JUnit, Git, Agile/Scrum, object-oriented design, unit testing, debugging

EDUCATION

University of Ottawa

Sep 2024 - Apr 2028

Honours Bachelor of Engineering, Computer Science | GPA: 3.6 / 4.0 | Ottawa, ON

- **Relevant coursework:** Machine Learning, Computer Vision, Algorithms and Data Structures, Operating Systems, Database Systems, Object-Oriented Software Engineering. Independent applied focus on sports computer vision, vision-language models, and AI-driven personalized education.
- **68 NeetCode DSA submissions** spanning arrays, graphs, dynamic programming, trees, and system design patterns, demonstrating consistent algorithmic practice applied to real-time sports analytics and adaptive learning platform architectures.
- **Eligible for Government of Canada security clearance**, a prerequisite credential for Ottawa-area defence, intelligence, and government contractor roles.

WORK EXPERIENCE

Founder and CEO, Mentrixia

Sep 2024 - Present

mentrixia.one | Ottawa, ON | Gamified EdTech SaaS: AI tutoring marketplace, adaptive practice engine, and competitive ranking system

- **Shipped a full-stack EdTech SaaS platform solo in 8 months** while enrolled full-time in second-year Computer Science, architecting a horizontally scalable multi-tenant system across Next.js 16 App Router, Python FastAPI, Supabase Postgres with 87 SQL migrations and full Row-Level Security, Drizzle ORM, Stripe Connect for automated tutor payouts, Resend transactional email, PostHog and Vercel Analytics for product telemetry, AWS S3 for file storage, and Sentry APM for full production observability.
- **Integrated Anthropic Claude API and Google Gemini 2.5 Flash** to build an adaptive AI quest generation and personalized practice engine that adjusts to individual learner progress in real time, demonstrating AI-driven personalization architecture for next-generation educational technology platforms.
- **Achieved 14/14 Playwright E2E tests and 44/44 Vitest unit tests passing**, shipping to Vercel with 11 scheduled cron jobs, edge proxy auth, Turnstile CAPTCHA, rate limiting, and Sentry monitoring, applying rigorous unit testing, debugging, and Agile/Scrum sprint cycles throughout the 8-month delivery.

- **Designed real-time skill duels, XP leveling, division leaderboards, and a clan system** using Supabase Realtime subscriptions for sub-second score propagation and Postgres FTS for fast in-app content search, driving measurable learner engagement and competitive retention.

Co-Founder, Solen AI

2025 - Present

Autonomous software platform that builds, deploys, monitors, and self-heals production applications from plain-language intent

- **Architected an end-to-end autonomous software factory** covering intent classification, architecture planning, LLM-driven code generation, Docker containerization, horizontal deployment, and real-time monitoring triggered from a single plain-English prompt, positioning Solen against Lovable, Bolt, and v0 in the infrastructure-for-AI-deployment category.
- **Engineered a self-healing closed-loop repair system** that monitors live deployments, diagnoses root cause using LLM reasoning, generates targeted fixes, rebuilds Docker images, and redeploys without human intervention, applying circuit-breaker patterns, multi-region failover logic, and an immutable audit trail covering every agent action and deployment event.

Software Engineer, AI Automation (Freelance)

May 2024 - Nov 2025

Upwork | Remote | 7 production clients across education, recruiting, and sports-adjacent SaaS verticals

- **Delivered 75% automation success rate uplift and 40% client ROI within 90 days** by engineering stateless Python FastAPI microservices with RabbitMQ async queues, applying horizontal scaling and circuit-breaker fault-tolerance patterns on AWS EC2 across 7 concurrent production environments, with multiple client systems remaining live in production today.
- **Increased qualified lead conversion by 35%** by architecting Python RAG workflows with LangChain and ChromaDB using TTL-cached embedding indexes and write-through cache invalidation for education-sector and recruiting clients, shipped via Jenkins CI/CD with zero-downtime deployment pipelines.
- **Reduced manual B2B workflow intervention by 45%** by building LLM-based classification modules with output guardrails, exponential-backoff retry logic, and Helm-packaged microservices deployed across 7 client environments, demonstrating full-cycle ownership from object-oriented system design to production rollout.

Software Developer Intern

May 2024 - Aug 2024

Hoopsalytics | Montreal, QC | Sports analytics startup building real-time computer vision infrastructure for professional basketball

- **Achieved 92% basketball-event detection accuracy at 30 FPS** by independently designing and deploying a computer vision inference system from scratch, selecting OpenCV and MediaPipe for sub-100ms CPU inference latency over GPU-dependent alternatives, deployed on live AWS EC2 streams across 200+ matches per season.
- **Reduced analyst review time by 30% per match** by building stateless Python FastAPI REST services with WebSocket streaming pushing live CV analytics into a React dashboard consumed by 8 professional analysts, eliminating all batch processing lag during live season and enabling real-time sports intelligence.
- **Sustained 99.9% pipeline uptime across a full professional basketball season** by architecting a staged ingestion-processing-inference backend with automated Bash deployment scripts enabling zero-downtime mid-season updates, applying debugging discipline and Git-managed version control throughout.

PROJECTS

NeuroDesk | Autonomous Multi-Agent Orchestration System

2026

FastAPI, PostgreSQL, Redis, Alembic, SQLAlchemy, Groq Llama 3.1, Anthropic Claude, Next.js, Tambo AI, SSE

- **Maintained sub-200ms response times under concurrent agentic workloads** by engineering NEXUS, a FastAPI orchestration engine parsing natural-language intent via Groq Llama 3.1 with Anthropic Claude fallback, routing work to specialized agents (ResearchAgent, CommunicationAgent, PurchaseAgent), and applying write-through Redis caching with async SQLAlchemy connection pooling.
- **Persisted an immutable audit trail to PostgreSQL via Alembic migrations** for every agent action, implementing human-in-the-loop budget enforcement and approval flows that intercept high-risk operations before execution, demonstrating the safety architecture required for autonomous AI in regulated educational and sports governance environments.

A.E.G.I.S. Garden | Real-Time SOC and Intelligent Decision Platform

2026

FastAPI, React, MongoDB Atlas, Snowflake, Google Gemini, OpenRouter, Solana, ElevenLabs, WebSocket, Docker

- **Processed incident triage from text, images, and structured data in real time** by integrating Google Gemini multimodal reasoning with strict JSON outputs and function-calling dispatch, layered over Snowflake Cortex for predictive demand forecasting via Snowpipe Streaming and MongoDB Atlas Change Streams for sub-second WebSocket state propagation.
- **Eliminated single-LLM-provider dependency** by integrating OpenRouter multi-model routing with automatic failover across providers based on latency, cost, and reasoning depth, implementing Solana Token Extensions for verifiable responder identities and ElevenLabs voice alerts for critical incident notification with full audit trail logging.

Chronos Cloud | Event-Driven Digital Twin | Best Snowflake API, Invest Ottawa 2026

Jan 2026

Python, FastAPI, NATS/JetStream, Docker, Kubernetes, Helm, PyTest, Jenkins

- **Achieved 99.99% message delivery reliability** by designing a distributed digital twin using NATS/JetStream pub-sub in Docker and Kubernetes with horizontal scaling and idempotent message processing, tracked via Git-managed Jenkins CI/CD. Won Best Use of Snowflake API at Invest Ottawa Hackathon 2026.

- **Reduced LLM-induced failures by 85%** by implementing a FastAPI control plane with circuit-breaker patterns, dead-letter queue fallback, and automated PyTest unit testing suites, sustaining 99.9% service continuity under simulated fault injection across distributed nodes.

B2B AI Workforce Optimizer | ML Scheduling and Predictive Analytics Platform

2025

FastAPI, Streamlit, DeepSeek AI, Plotly 3D, SQLite, Twilio, Python

- **Reduced staffing decision errors** by building a real-time demand prediction engine using DeepSeek AI with write-optimized FastAPI ingestion and constraint-aware scheduling, applying the same predictive analytics architecture used in sports roster optimization and educational cohort scheduling.
- **Delivered real-time workforce command visibility** by designing an interactive Streamlit dashboard with Plotly 3D trend visualizations and Twilio SMS alerting for scheduling anomalies and demand spikes, enabling operations managers to act on staffing signals without active dashboard access.

Distributed Systems Developer | University of Ottawa AI+ Society

Oct 2025 - Jan 2026

Ottawa, ON | Event-driven backend infrastructure for a 3-person Agile team

- **Reduced event processing latency by 30%** by replacing a synchronous polling architecture with RabbitMQ async message queues, decoupling producers from consumers and enabling backpressure-controlled throughput scaling across the society's internal tooling infrastructure.
- **Improved on-time feature delivery by 30% across 6 Agile sprint cycles** by implementing GitHub Actions CI/CD with automated unit testing gates and tagged release management, contributing as a backend developer using object-oriented design patterns in a collaborative Git-managed repository.

CERTIFICATIONS

AWS Fundamentals of Machine Learning and Artificial Intelligence | *Amazon Web Services* | Oct 2025

Intro to Machine Learning, scikit-learn, Random Forest, model evaluation | *Kaggle* | Nov 2025

Practical GitHub Actions | *LinkedIn Learning* | Feb 2026

Introduction to Artificial Intelligence | *LinkedIn Learning* | 2025

Docker and DevOps Fundamentals | *In Progress* | 2026

AWS Cloud Practitioner | *Amazon Web Services* | 2026

AWARDS AND RECOGNITION

- **Best Use of Snowflake API**, Invest Ottawa Hackathon 2026, Chronos Cloud distributed digital twin platform.
- **Faculty of Science Achievement Award**, University of Ottawa 2024-2025, awarded to top-performing Faculty of Science students.
- **68 verified NeetCode DSA submissions** across arrays, graphs, dynamic programming, trees, and system design patterns.
- **GitHub Pull Shark and YOLO Achievement Badges**, consistent open-source commit activity and pull request contributions.